RE501 Class 3: Probability Distributions

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In Class 2 (Section 29) we defined a random variable $X : \Omega \to \mathbb{R}$ and its probability mass function $\Pr_X(x) = \Pr(X = x)$. In Class 2 (Sections 32 and 33) we learnt to summarise a distribution by its mean $\bar{X}$ and variance σ_X^2. In this class we meet the binomial and Poisson distributions for counts, the cumulative distribution function (CDF) and probability density function (PDF), and the Gaussian and exponential distributions for continuous quantities.

Chapter 7: Discrete distributions

35 Binomial distribution: N tosses of a biased coin

Take a *biased* coin: it shows Head with probability p and Tail with probability $1 - p$, where p is some number in $[0, 1]$. (A fair coin is the special case $p = 1/2$.) Toss this coin N times. The tosses do not influence each other, so they are *independent* (Class 2, Section 27).

Let the random variable X be the *number of Heads* in the N tosses. X can take the values $0, 1, 2, \ldots, N$. We want $\Pr_X(x) = \Pr(X = x)$.

Step 1: three tosses, written out

For $N = 3$ the sample space has $2^3 = 8$ outcomes. Because the tosses are independent, the probability of a sequence is the product of the probabilities of its tosses, e.g. $\Pr(\{HTH\}) = p \cdot (1 - p) \cdot p = p^2(1 - p)$.

Value x	Preimage $X^{-1}(x)$	Probability of *each* outcome	$\Pr_X(x)$
0	$\{TTT\}$	$(1-p)^3$	$1 \cdot (1-p)^3$
1	$\{HTT, THT, TTH\}$	$p\,(1-p)^2$	$3\,p\,(1-p)^2$
2	$\{HHT, HTH, THH\}$	$p^2(1-p)$	$3\,p^2(1-p)$
3	$\{HHH\}$	p^3	$1 \cdot p^3$

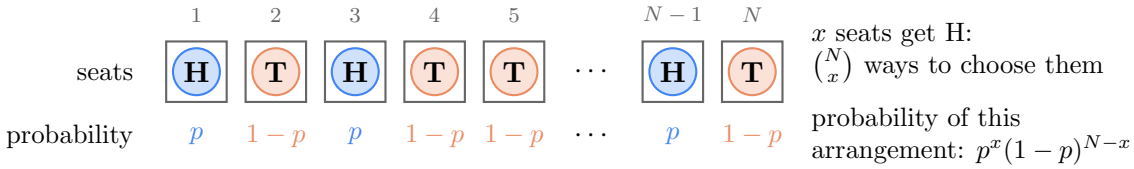
Two things happen in every row:

1. Every outcome with x Heads has the *same* probability, $p^x(1 - p)^{3-x}$: x factors of p and $3 - x$ factors of $(1 - p)$, in some order.
2. The number of such outcomes is the number of ways to choose *which* x of the 3 tosses are Heads: $\binom{3}{x} = 1, 3, 3, 1$.

The outcomes in a row are mutually exclusive, so their probabilities add (Class 2, Section 25): $\Pr_X(x) = (\text{count}) \times (\text{probability of each})$.

Step 2: any number of tosses

The same two observations hold for any N. Picture the N tosses as N empty seats in a row, one seat per toss. An outcome with exactly x Heads is a way of placing x coins showing H on x of the seats; the remaining $N - x$ seats get coins showing T.



1. **Counting the arrangements.** Placing the x Heads means choosing which x of the N seats they occupy; the order in which we choose does not matter. This is exactly the number of x-element subsets of an N-element set, $\binom{N}{x}$ (Class 1, Section 8).
2. **Probability of one arrangement.** The tosses are independent, so we multiply the probabilities seat by seat: each of the x seats with H contributes a factor p, and each of the $N - x$ seats with T contributes $1 - p$. Every arrangement with x Heads therefore has the same probability, $p^x(1-p)^{N-x}$.
3. **Adding up.** Different arrangements are mutually exclusive outcomes, so their probabilities add: $\binom{N}{x}$ arrangements, each of probability $p^x(1-p)^{N-x}$.

Hence

Binomial distribution. $X \sim B(N, p)$ means

$$\Pr_X(x) = \binom{N}{x} p^x (1-p)^{N-x}, \qquad x = 0, 1, 2, \ldots, N.$$

The two numbers N (number of tosses) and p (probability of Head in one toss) are the **parameters** of the distribution. (Some books write $\mathrm{Bin}(N, p)$ or $\mathrm{Binom}(N, p)$; the parameters are always listed in the order N, then p.)

Check: the probabilities add to 1. In Class 1 (Section 8) we used the binomial expansion $(1 + x)^n = \sum_k \binom{n}{k} x^k$ to count the elements of a power set. The general form of the same expansion is $(a + b)^N = \sum_{x=0}^{N} \binom{N}{x} a^x b^{N-x}$. With $a = p$ and $b = 1 - p$:

$$\sum_{x=0}^{N} \Pr_X(x) = \sum_{x=0}^{N} \binom{N}{x} p^x (1-p)^{N-x} = \big(p + (1-p)\big)^N = 1^N = 1.$$

This is where the distribution gets its name: its probabilities are the terms of a binomial expansion.

Example. A coin with $p = 0.3$ tossed $N = 5$ times gives exactly 2 Heads with probability $\binom{5}{2}(0.3)^2(0.7)^3 = 0.3087$.

36 The true/false random variable

Nothing in Section 35 used the fact that we had a coin. We only used three things:

1. each repetition (**trial**) has two results: "yes" or "no";
2. the probability of "yes" is the same number p in every trial;
3. the trials are independent.

So take *any* random experiment and *any* event $E \in \sigma(\Omega)$ we care about, with $\Pr(E) = p$. Define a random variable for one trial in a true/false manner:

$$Y(\omega) = \begin{cases} 1 & \text{if } \omega \in E \quad (\text{“true”: } E \text{ occurred}), \\ 0 & \text{if } \omega \notin E \quad (\text{“false”: } E \text{ did not occur}). \end{cases}$$

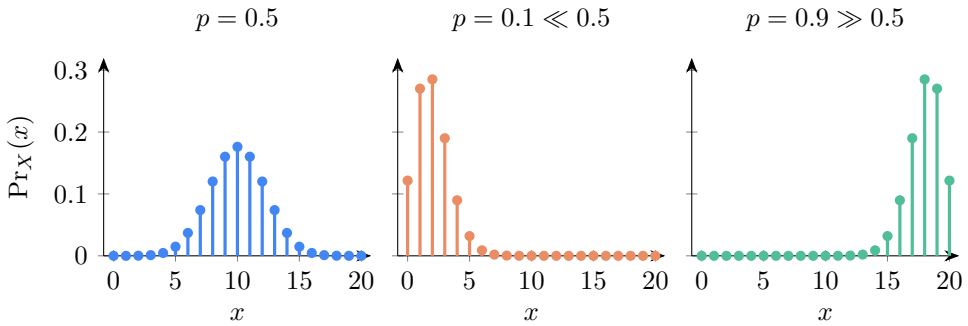
Then $\Pr(Y = 1) = p$ and $\Pr(Y = 0) = 1 - p$. This is exactly a biased coin, with “Head” renamed “E occurred”. (A single true/false trial is called a **Bernoulli trial**.)

Repeat the experiment N times independently, and let $Y_1, Y_2, \ldots, Y_N$ be the true/false values. The total $X = Y_1 + Y_2 + \cdots + Y_N$ counts the number of trials in which E occurred, and by Section 35, $X \sim B(N, p)$.

Examples. Roll a die and ask for a six ($p = 1/6$); inspect items from a production line and ask whether each is defective ($p =$ defect rate); send bits over a noisy channel and ask whether each is flipped ($p =$ bit error rate). In each case, the number of “true” results in N independent trials follows the binomial distribution.

37 Shape of the binomial distribution

The plots below show $\Pr_X(x)$ for $N = 20$ and three values of p.



- $p \approx 0.5$: the distribution is *symmetric*, with its peak at the middle, $x \approx N/2 = 10$. It has a bell shape; we will meet this shape again in Section 46.
- $p \ll 0.5$: “true” is rare, so small counts are likely. The peak sits near $x = 0$ and the distribution has a long tail to the right.
- $p \gg 0.5$: the mirror image. The peak sits near $x = N$ with a long tail to the left. This is the same as the previous case with the roles of “true” and “false” swapped: $B(20, 0.9)$ for the number of Heads is $B(20, 0.1)$ for the number of Tails.

In every case the peak is near Np.

Mean and variance

Using the definitions of Class 2 (Sections 32 and 33), one can show (we do not prove it here) that for $X \sim B(N, p)$

$$\bar{X} = Np, \qquad \sigma_X^2 = Np(1-p), \qquad \sigma_X = \sqrt{Np(1-p)}.$$

The mean agrees with intuition: toss a coin with $p = 0.3$ a hundred times and you expect about 30 Heads. The variance is largest at $p = 0.5$ and goes to zero as $p \to 0$ or $p \to 1$: a coin that always shows Head (or always Tail) has no spread at all.

38 Poisson distribution: the limit of many rare trials

Many counting problems have a huge number of trials, each with a tiny probability of "true": the number of nuclei that decay in a sample, the number of calls arriving at an exchange in a minute, the number of defects on a long cable. Here N is very large and p is very small, but the mean number of "true" results, Np, is a moderate number.

It can be shown (we do not prove it here) that the binomial distribution approaches a new distribution under the following conditions:

1. the number of trials is very large, $N \to \infty$;
2. the probability of "true" in each trial is very small, $p \to 0$;
3. their product, the mean, stays a fixed finite number: $Np = \bar{X} = \lambda$.

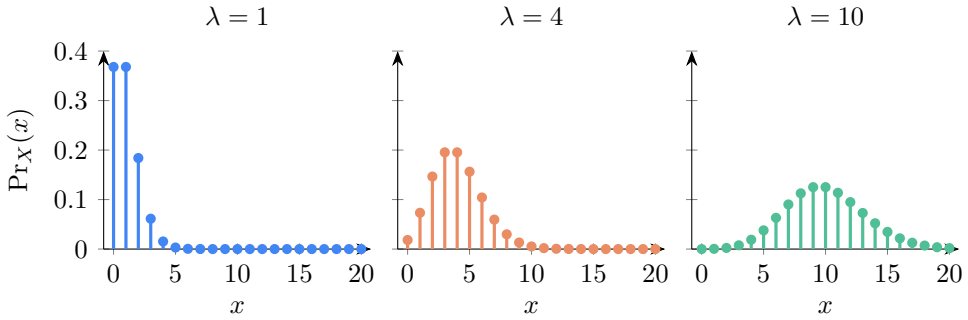
Poisson distribution. Under these conditions $X \sim B(N, p)$ becomes $X \sim \text{Poisson}(\lambda)$, with

$$\Pr_X(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \qquad x = 0, 1, 2, \ldots$$

It has a single parameter $\lambda > 0$. Unlike the binomial, x has no upper limit.

Check: the probabilities add to 1. Since $e^\lambda = \sum_{x=0}^{\infty} \lambda^x / x!$, we get $\sum_x \Pr_X(x) = e^{-\lambda} e^\lambda = 1$.

39 Shape of the Poisson distribution



- Small λ: the distribution is piled up near $x = 0$ with a tail to the right, like the binomial with $p \ll 0.5$ (which is where it came from).
- As λ grows, the peak moves to $x \approx \lambda$, the distribution spreads out, and it becomes more symmetric and bell shaped.

Mean and variance

We state without proof that for $X \sim \text{Poisson}(\lambda)$

$$\bar{X} = \lambda, \qquad \sigma_X^2 = \lambda.$$

For the Poisson distribution the mean and the variance are equal.

This is consistent with the binomial result of Section 37: $\bar{X} = Np = \lambda$, and $\sigma_X^2 = Np(1-p) = \lambda(1-p) \to \lambda$ as $p \to 0$.

Why this matters. Since $\sigma_X = \sqrt{\lambda}$, the relative fluctuation of a count is $1/\sqrt{\lambda}$: about 10% at $\lambda = 100$, 1% at $\lambda = 10\,000$. This is the origin of the $\pm\sqrt{N}$ error bars on counts (Section 46).

40 Example: radioactive decay

Decay of a single nucleus is a memoryless process. An unstable nucleus does not age. However long it has already existed, the probability that it decays in the next short time δt is always the same, and it is proportional to δt:

$$\Pr\big(\text{decays in the next } \delta t \,\big|\, \text{has survived so far}\big) = \lambda\,\delta t.$$

The constant λ (units 1/time) depends only on the isotope. In Section 48 we will show that λ is the **decay constant** of the isotope, related to its half-life by

$$\lambda = \frac{\ln 2}{T_{1/2}},$$

and that the probability that a nucleus decays within an observation time δt (not necessarily short) is

$$p = 1 - e^{-\lambda\,\delta t} = 1 - 2^{-\delta t/T_{1/2}}.$$

Binomial or Poisson? The role of δt. Take a sample of N nuclei, observe it for a time δt, and count the decays. Apply the true/false idea of Section 36: each nucleus is one trial, "true" means "decayed within δt", with probability p. So the number of decays is $X \sim B(N, p)$. Which distribution we use in practice depends on how δt compares with $T_{1/2}$:

- δt **comparable to** $T_{1/2}$**.** If $T_{1/2} = 1$ s and we look at the nucleus after 1 s, the probability that it has decayed is $p = 1 - 2^{-1} = \frac{1}{2}$. This p is not small, so the Poisson limit (Section 38) does not apply. Such short-lived nuclides are usually present in small numbers, and we must use the **binomial** distribution.
- δt **much shorter than** $T_{1/2}$**.** If $T_{1/2}$ is thousands of years and we observe for 1 s, the probability that a given nucleus decays is negligible, $p \approx \lambda\,\delta t$. If the number of nuclei is large, $N \to \infty$ and $p \to 0$ with Np finite, and we use the **Poisson** distribution with mean $\bar{X} = Np$.

Both descriptions assume that the decay of one nucleus does not influence the decay of any other nucleus (independent trials, Section 36).

Radioactive decay satisfies this very well. A counter-example is **neutron-induced fission** in a fissile material such as uranium-235: each fission releases two or three neutrons, and these neutrons can cause fission in neighbouring nuclei. The fission events are then *not* independent; one event raises the chance of others (a chain reaction). The number of fissions in a time bin then follows neither the binomial nor the Poisson distribution, and the counts fluctuate more than Poisson statistics would predict (variance larger than the mean).

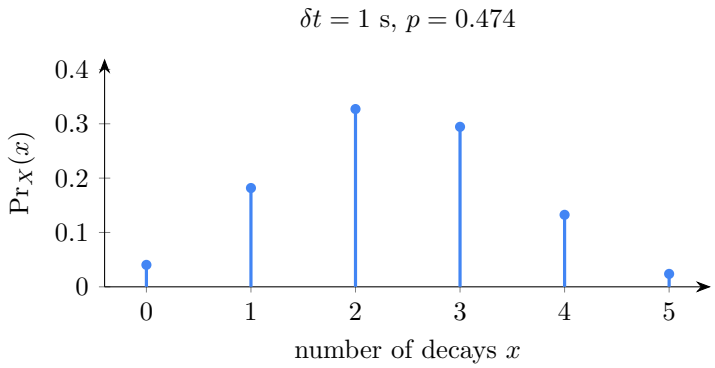
Few nuclides, short half-life: sodium-26

Sodium-26 has a half-life $T_{1/2} = 1.08$ s. Suppose we have just $N = 5$ such nuclei and observe them for $\delta t = 1$ s. The probability that one nucleus decays in this window is

$$p = 1 - 2^{-1/1.08} \approx 0.474,$$

so the number of decays is $X \sim B(5, 0.474)$:

x	0	1	2	3	4	5
$\Pr_X(x)$	0.040	0.182	**0.327**	0.294	0.133	0.024



In a 1 s window the most likely result is 2 decays (about 33%); 4 decays has probability only about 13%. A longer window would raise p and shift the peak to more decays: the window δt matters because it sets p.

The Poisson distribution would be wrong here. With the same mean $Np = 2.37$, it gives a probability of about 3% to *six or more* decays, which is impossible with only five nuclei.

Many nuclides, long half-life: one gram of carbon-14

Carbon-14 has $T_{1/2} = 5730$ years $= 5730 \times 3.156 \times 10^7$ s $\approx 1.81 \times 10^{11}$ s, so

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{1.81 \times 10^{11}\ \text{s}} \approx 3.83 \times 10^{-12}\ \text{s}^{-1}.$$

One gram of carbon-14 (molar mass 14 g/mol) contains

$$N = \frac{6.022 \times 10^{23}}{14} \approx 4.30 \times 10^{22}\ \text{nuclei}.$$

In a $\delta t = 1$ s window, the probability that a given nucleus decays is

$$p = 1 - e^{-\lambda\,\delta t} \approx \lambda\,\delta t \approx 3.83 \times 10^{-12},$$

utterly negligible, while N is enormous. This is exactly the Poisson limit: the number of decays per second is Poisson($\bar{X}$) with

$$\bar{X} = Np = N\lambda\,\delta t \approx 4.30 \times 10^{22} \times 3.83 \times 10^{-12} \approx 1.65 \times 10^{11}\ \text{decays per second}.$$

Its standard deviation is $\sqrt{\bar{X}} \approx 4.1 \times 10^5$, a relative fluctuation of only 2.5×10^{-6} (Section 39). Also, in an hour of counting only a fraction $\lambda \times 3600$ s $\approx 1.4 \times 10^{-8}$ of the nuclei decay, so the mean stays the same from bin to bin.

Activity. The expected number of decays per unit time, $A = N\lambda$, is called the **activity**. Its SI unit is the becquerel (1 Bq = 1 decay per second); the traditional unit is the curie, 1 Ci $= 3.7 \times 10^{10}$ Bq. For carbon-14,

$$A = N\lambda \approx 1.65 \times 10^{11}\ \text{Bq per gram} = \frac{1.65 \times 10^{11}}{3.7 \times 10^{10}}\ \text{Ci/g} \approx 4.46\ \text{Ci/g}.$$

Chapter 8: Cumulative distribution and probability density

41 Cumulative distribution function

So far we described a random variable by the probability of each single value. A second description, which works equally well for discrete and continuous random variables, uses the probability of *everything up to* a value.

Cumulative distribution function (CDF). For a random variable X and any real number y,

$$F_X(y) = \Pr_X\left((-\infty, y]\right) = \Pr(X \le y).$$

Here $(-\infty, y]$ is an event in $\sigma(\mathbb{R})$, so $\Pr_X$ of it is defined (Class 2, Section 29).

Properties.

1. F_X **is an increasing (non-decreasing) function:** if $y_1 < y_2$ then $F_X(y_1) \le F_X(y_2)$.

   Reason. The interval $(-\infty, y_2]$ is the union of the mutually exclusive sets $(-\infty, y_1]$ and $(y_1, y_2]$. By the addition axiom,

   $$F_X(y_2) = F_X(y_1) + \Pr_X\left((y_1, y_2]\right) \ge F_X(y_1),$$

   since a probability is never negative.

2. $F_X(-\infty) = 0$, meaning $\lim_{y\to-\infty} F_X(y) = 0$: as y decreases, $(-\infty, y]$ shrinks towards the empty set, and $\Pr_X(\emptyset) = 0$.

3. $F_X(\infty) = 1$, meaning $\lim_{y\to\infty} F_X(y) = 1$: as y increases, $(-\infty, y]$ grows towards all of $\mathbb{R}$, and $\Pr_X(\mathbb{R}) = 1$.

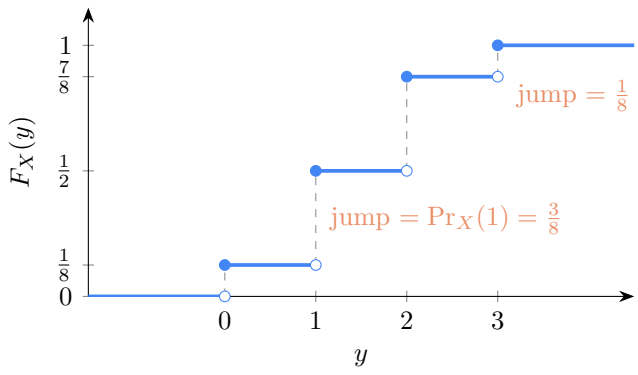
So every CDF climbs from 0 to 1 and never goes down.

Probability of an interval. The equation in property 1 gives a very useful rule:

$$\Pr(y_1 < X \le y_2) = F_X(y_2) - F_X(y_1).$$

42 CDF of a discrete random variable

Let X be the number of Heads in three tosses of a fair coin, $X \sim B(3, 0.5)$. From Section 35, $\Pr_X(0) = \frac{1}{8}$, $\Pr_X(1) = \frac{3}{8}$, $\Pr_X(2) = \frac{3}{8}$, $\Pr_X(3) = \frac{1}{8}$. Adding up everything at or below y, $F_X(y)$ is 0 for $y < 0$ and then steps up to $\frac{1}{8}$ at $y = 0$, $\frac{1}{2}$ at $y = 1$, $\frac{7}{8}$ at $y = 2$ and 1 at $y = 3$:

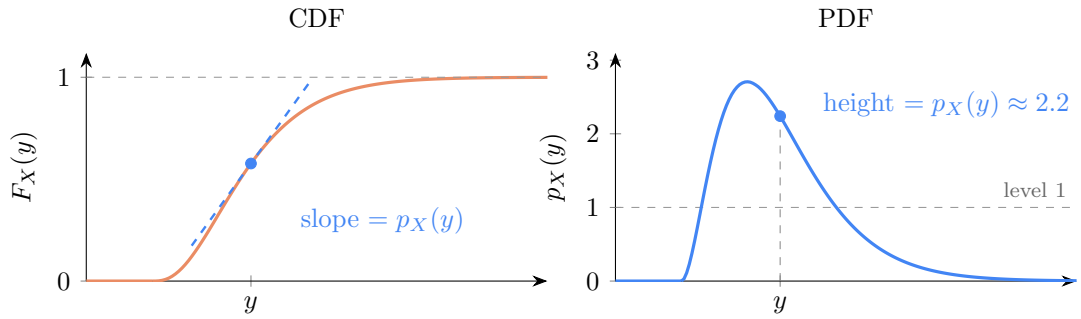


- The CDF of a discrete random variable is a *staircase*. It is flat between the possible values and **jumps** at each possible value x_i.
- The **size of the jump** at x_i is exactly $\Pr_X(x_i)$. So the CDF contains the same information as the probability mass function.
- F_X is **discontinuous** at every possible value. At the jump the CDF takes the *upper* value (filled dot), because $\Pr(X \le y)$ includes $X = y$. For instance $F_X(1) = \frac{1}{2}$, while just below 1, $F_X(0.999) = \frac{1}{8}$.

43 Continuous random variables and the PDF

Many quantities vary continuously: the voltage across a resistor, the lifetime of a bearing, the time until the next decay. A random variable whose CDF $F_X(y)$ is a *continuous* function of y (no jumps) is called a **continuous random variable**. Since a jump in F_X at x has size $\Pr_X(x)$, no jumps means $\Pr(X = x) = 0$ for every single value x. So for a continuous random variable we ask for the probability of an *interval*, not of a point.

An illustrative CDF. The left figure shows a typical CDF of a continuous random variable, rising smoothly from 0 to 1. The dashed line is the tangent at a point y; its slope tells us how fast probability accumulates near y. The right figure plots this slope at every point: the PDF defined below. **Notice that the CDF never exceeds 1, but the PDF does:** here the CDF climbs steeply, so its slope, the PDF, reaches about 2.7.



Probability density function (PDF). The PDF of a continuous random variable X is the slope of its CDF:

$$p_X(x) = \left.\frac{dF_X(y)}{dy}\right|_{y=x}.$$

Integrating both sides. Integrate the definition from $-\infty$ to y:

$$\int_{-\infty}^{y} p_X(x)\,dx = F_X(y) - F_X(-\infty) = F_X(y), \qquad \text{since } F_X(-\infty) = 0.$$

$$F_X(y) = \int_{-\infty}^{y} p_X(x)\,dx.$$

Since F_X never decreases, its slope is never negative: $p_X(x) \ge 0$. From this one relation, the other properties of the PDF follow one by one:

- **Normalisation.** Let $y \to \infty$ and use $F_X(\infty) = 1$:

$$\int_{-\infty}^{\infty} p_X(x)\,dx = 1. \qquad \text{The total area under the PDF is 1.}$$

- **Area from $-\infty$ to y.** By the boxed relation, the area under the PDF to the left of y is the probability of everything up to y:

$$\int_{-\infty}^{y} p_X(x)\,dx = F_X(y) = \Pr(-\infty < X \le y).$$

- **Area from a to b.** For $a < b$, using the interval rule of Section 41,

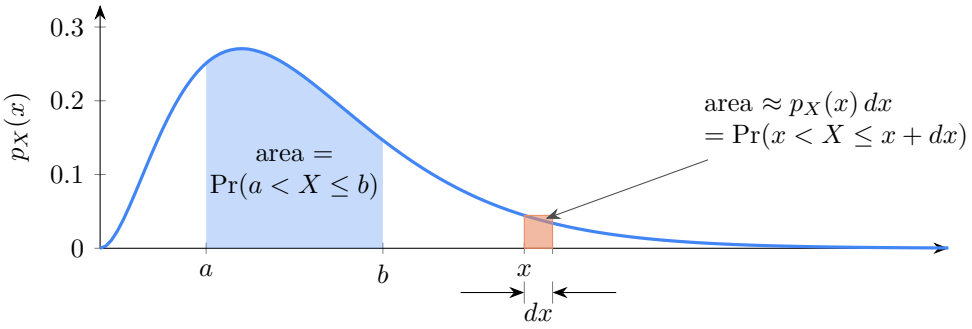
$$\Pr(a < X \le b) = F_X(b) - F_X(a) = \int_{-\infty}^{b} p_X(x)\,dx - \int_{-\infty}^{a} p_X(x)\,dx = \int_{a}^{b} p_X(x)\,dx.$$

  The probability that X lies between a and b is the area under the PDF between $x = a$ and $x = b$.

- **Small intervals.** Take $b = x + dx$ with dx very small. Over such a short interval p_X hardly changes, so the area is almost a thin rectangle of height $p_X(x)$ and width dx:

$$\Pr(x < X \le x + dx) = \int_{x}^{x+dx} p_X(t)\,dt \approx p_X(x)\,dx.$$

  This also follows directly from the slope: $F_X(x + dx) - F_X(x) \approx F_X'(x)\,dx = p_X(x)\,dx$. It becomes exact as $dx \to 0$.



This is the reason for the name *density*: $p_X(x)$ is not itself a probability (it can be larger than 1) but a probability *per unit* of x, in the same way that mass density is mass per unit volume. If x is measured in seconds, $p_X(x)$ has units of s^{-1}. ***Though probability is a dimensionless pure number, the probability density may not be a dimensionless quantity.*** A single point has probability zero, since $p_X(x)\,dx \to 0$ as $dx \to 0$.

44 Mean and variance from the PDF

In Class 2 (Sections 32 and 33) the mean and variance of a discrete random variable were sums over the possible values, each weighted by its probability $\Pr_X(x_i)$. For a continuous random variable, chop the x-axis into small pieces of width dx. The piece at x carries probability $p_X(x)\,dx$ (Section 43). Replacing $x_i \to x$, $\Pr_X(x_i) \to p_X(x)\,dx$ and $\sum_i \to \int_{-\infty}^{\infty}$ turns the sums into integrals:

Mean:	$\bar{X} = E[X] = \displaystyle\int_{-\infty}^{\infty} x\,p_X(x)\,dx,$
Variance:	$\sigma_X^2 = \displaystyle\int_{-\infty}^{\infty} (x - \bar{X})^2\,p_X(x)\,dx = E[X^2] - \bar{X}^2, \qquad E[X^2] = \int_{-\infty}^{\infty} x^2\,p_X(x)\,dx.$

The relation $\sigma_X^2 = E[X^2] - \bar{X}^2$ is proved exactly as in Class 2 (Section 34.2), with sums replaced by integrals. As before, the n-th raw moment is $E[X^n] = \int x^n p_X(x)\,dx$.

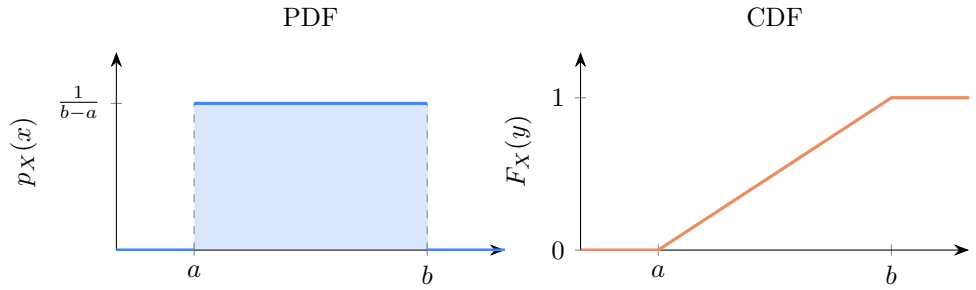
Chapter 9: Continuous distributions

45 Uniform distribution

Uniform distribution. $X \sim \mathcal{U}(a, b)$, with $a < b$, means

$$p_X(x) = \begin{cases} \dfrac{1}{b-a} & a \le x \le b, \\ 0 & \text{otherwise.} \end{cases}$$

Every value between a and b is equally likely; values outside are impossible. The height $1/(b-a)$ is fixed by normalisation: a rectangle of width $b-a$ must have area 1. Integrating, the CDF rises in a straight line from 0 at $y = a$ to 1 at $y = b$: $F_X(y) = (y-a)/(b-a)$ for $a \le y \le b$.



Mean and variance. Here the integrals are elementary:

$$\bar{X} = \int_a^b \frac{x}{b-a}\,dx = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}, \qquad E[X^2] = \int_a^b \frac{x^2}{b-a}\,dx = \frac{b^3-a^3}{3(b-a)} = \frac{a^2+ab+b^2}{3},$$

$$\sigma_X^2 = E[X^2] - \bar{X}^2 = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}.$$

The mean is the midpoint, as expected. **Example:** the rounding error of a digital meter is uniform on $[-\frac{1}{2}, \frac{1}{2}]$ of the last digit, so its standard deviation is $1/\sqrt{12} \approx 0.29$ of the last digit.

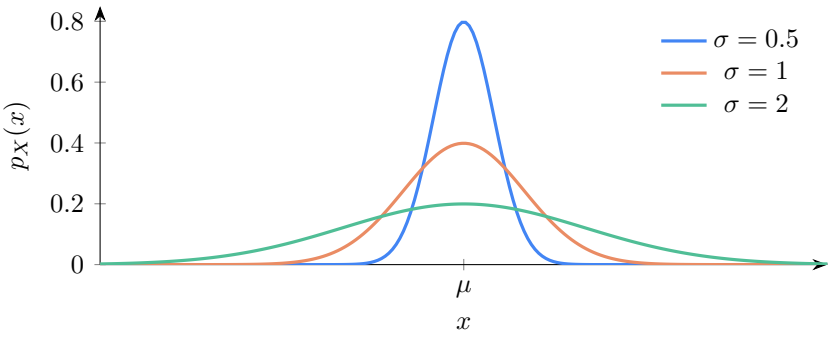
46 Gaussian distribution

Gaussian (normal) distribution. $X \sim \mathcal{N}(\mu, \sigma^2)$ means

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad -\infty < x < \infty.$$

It has two parameters, μ and $\sigma^2 > 0$.

The curve is the familiar symmetric bell. It peaks at $x = \mu$, and σ sets its width: a small σ gives a tall, narrow bell, a large σ a low, wide one (the area is always 1).



Where it comes from. Look back at the binomial distribution with $p \approx 0.5$ in Section 37: it already has this bell shape. In fact, the Gaussian can be derived from $B(N, p)$ (we do not do this here) when

1. the number of trials is large, $N \to \infty$;
2. p stays fixed and is not close to 0 or 1, so that both Np and $N(1-p)$ are large (in practice $\gtrsim 10$). The approximation works best for $p \approx 0.5$, where the binomial is already symmetric.

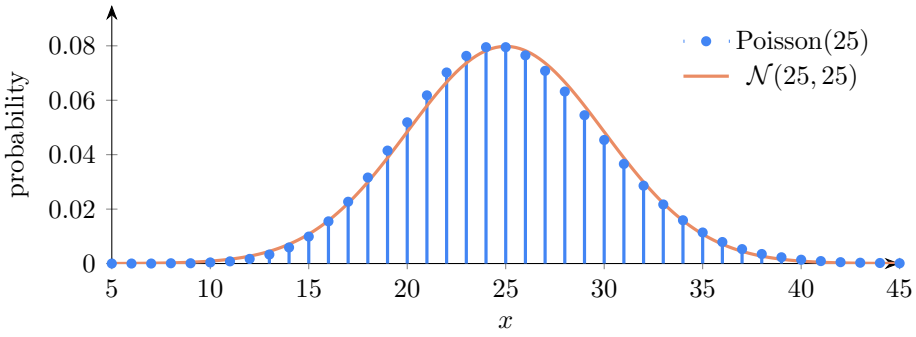
Compare with the Poisson limit of Section 38, where $p \to 0$ while Np stays finite. The resulting Gaussian has

$$\mu = Np, \qquad \sigma^2 = Np(1-p).$$

Poisson also becomes Gaussian. The same happens to the Poisson distribution when its mean is large. Look at $\lambda = 10$ in Section 39: it is already nearly a bell. For large λ (in practice $\lambda \gtrsim 20$),

$$\text{Poisson}(\lambda) \approx \mathcal{N}(\lambda, \lambda),$$

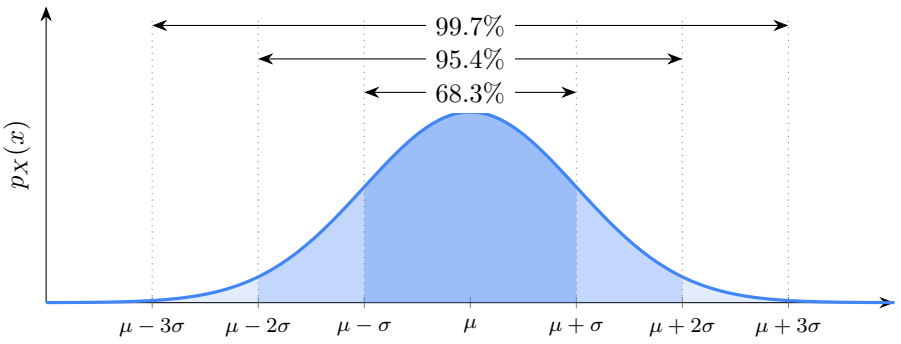
a Gaussian whose mean and variance are both λ (Section 39). Below, Poisson(25) (stems) is compared with $\mathcal{N}(25, 25)$ (curve):



The 68–95–99.7 rule. For any Gaussian, the probability of landing within one, two or three standard deviations of the mean is always the same:

$$\Pr(|X-\mu| \le \sigma) \approx 0.683, \qquad \Pr(|X-\mu| \le 2\sigma) \approx 0.954, \qquad \Pr(|X-\mu| \le 3\sigma) \approx 0.997.$$

These numbers are the areas $\int_{\mu-k\sigma}^{\mu+k\sigma} p_X(x)\,dx$ for $k = 1, 2, 3$. The Gaussian CDF has no formula in elementary functions, so they are found numerically (or read from a table). In words: about two thirds of measurements fall within $\pm\sigma$, about 95% within $\pm 2\sigma$, and a result more than 3σ from the mean happens only about 3 times in 1000.



Why counts carry $\pm\sqrt{N}$ error bars

Suppose a counter records N counts in one time bin (Section 40). The count is Poisson with some unknown mean λ, and our best estimate of λ is the count itself, $\lambda \approx N$. When N is large:

1. the count is approximately Gaussian, $\mathcal{N}(\lambda, \lambda)$ (Poisson becomes Gaussian);
2. its standard deviation is $\sigma = \sqrt{\lambda} \approx \sqrt{N}$ (Poisson mean = variance);
3. by the 68–95–99.7 rule, a single count lands within $\pm\sigma$ of λ about 68% of the time.

So a measured count is reported as

$$N \pm \sqrt{N},$$

meaning the true mean lies in this range with about 68% confidence ($N \pm 2\sqrt{N}$ for about 95%). For example, 400 counts is reported as 400 ± 20, a 5% uncertainty; to reach 1% we need $N = 10\,000$ counts. All three facts are needed: without the Gaussian limit, $\pm\sqrt{N}$ would not come with the 68% meaning.

47 Integrating the Gaussian: the Gamma function

To find the mean and variance we need integrals of the form $\int x^n e^{-(x-\mu)^2/2\sigma^2} dx$. A change of variable turns all of them into one standard integral.

Gamma function. For $n > 0$,

$$\Gamma(n) = \int_0^\infty z^{\,n-1} e^{-z}\, dz.$$

Properties.

- $\Gamma(n) = (n-1)\,\Gamma(n-1)$ (integrate by parts);
- $\Gamma(1) = 1$, so $\Gamma(n) = (n-1)!$ when n is a positive integer;
- $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, and hence $\Gamma(\frac{3}{2}) = \frac{1}{2}\Gamma(\frac{1}{2}) = \frac{\sqrt{\pi}}{2}$.

The substitution. The Gaussian is symmetric about $x = \mu$, so we integrate over $x \geq \mu$ and double where needed. For $x \geq \mu$ put

$$z = \frac{(x-\mu)^2}{2\sigma^2}, \qquad \text{so} \qquad x - \mu = \sigma\sqrt{2z}, \qquad dx = \frac{\sigma}{\sqrt{2z}}\, dz,$$

and z runs from 0 to ∞ as x runs from μ to ∞.

Mean and variance of the Gaussian

Normalisation.

$$\int_{-\infty}^{\infty} p_X(x)\, dx = \frac{2}{\sqrt{2\pi}\,\sigma} \int_\mu^\infty e^{-(x-\mu)^2/2\sigma^2}\, dx = \frac{2}{\sqrt{2\pi}\,\sigma} \cdot \frac{\sigma}{\sqrt{2}} \int_0^\infty z^{-1/2} e^{-z}\, dz = \frac{1}{\sqrt{\pi}}\,\Gamma(\tfrac{1}{2}) = 1.$$

Mean. Write $x = (x-\mu)+\mu$:

$$\bar{X} = \int_{-\infty}^{\infty} (x-\mu)\, p_X(x)\, dx + \mu \int_{-\infty}^{\infty} p_X(x)\, dx = 0 + \mu = \mu.$$

The first integral is zero because $(x-\mu)\,p_X(x)$ is odd about $x=\mu$: its values at $\mu-u$ and $\mu+u$ cancel.

Second moment. First the second central moment, using $(x-\mu)^2 = 2\sigma^2 z$:

$$\begin{aligned} E[(X-\mu)^2] &= \frac{2}{\sqrt{2\pi}\,\sigma} \int_{\mu}^{\infty} (x-\mu)^2 e^{-(x-\mu)^2/2\sigma^2}\, dx = \frac{2}{\sqrt{2\pi}\,\sigma} \int_0^{\infty} 2\sigma^2 z\, e^{-z}\, \frac{\sigma}{\sqrt{2z}}\, dz \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} z^{1/2} e^{-z}\, dz = \frac{2\sigma^2}{\sqrt{\pi}}\, \Gamma(\tfrac{3}{2}) = \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \sigma^2. \end{aligned}$$

Then, expanding $x^2 = (x-\mu)^2 + 2\mu(x-\mu) + \mu^2$,

$$E[X^2] = E[(X-\mu)^2] + 2\mu\, E[X-\mu] + \mu^2 = \sigma^2 + 0 + \mu^2 = \sigma^2 + \mu^2.$$

Variance.

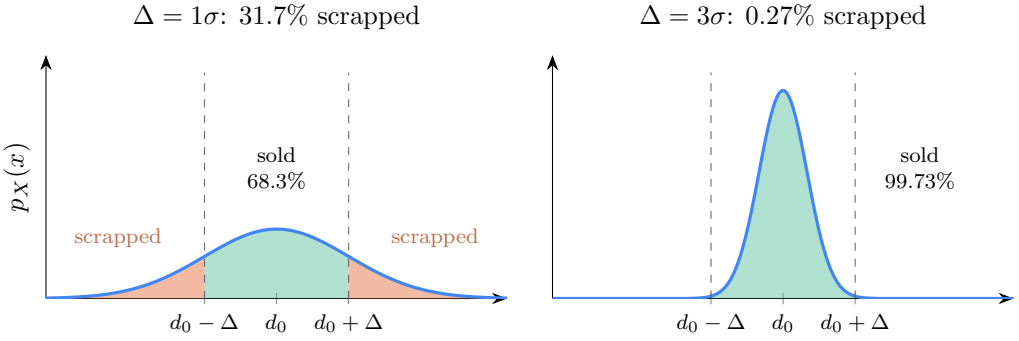
$$\mathrm{Var}(X) = \sigma_X^2 = E[X^2] - \bar{X}^2 = (\sigma^2+\mu^2) - \mu^2 = \sigma^2.$$

For $X \sim \mathcal{N}(\mu, \sigma^2)$: $\quad \bar{X} = \mu, \quad E[X^2] = \sigma^2 + \mu^2, \quad \sigma_X^2 = \sigma^2.$
The two parameters of the Gaussian are exactly its mean and its variance.

Six sigma: from quality control to scientific discovery

A rod manufacturer. A manufacturer makes steel rods of a nominal diameter d_0. The buyer accepts a rod only if its diameter lies within the tolerance $d_0 \pm \Delta$. No machine makes every rod identical: the weather (temperature, humidity), the batch of raw steel supplied, its chemistry and the wear of the tools all change a little from rod to rod. So the diameter X of a rod is a random variable. Since it is the sum of many small, independent causes, we model it as a Gaussian, $X \sim \mathcal{N}(\mu, \sigma^2)$, with the machine set to $\mu = d_0$. Every rod is measured after production. If it is within tolerance it is sent to the seller; otherwise it is destroyed (scrapped).

The fraction of good rods, $\Pr(d_0 - \Delta \le X \le d_0 + \Delta)$, depends only on how many σ fit into the tolerance Δ: this is the 68–95–99.7 rule of Section 46. Below, two machines work to the same tolerance. The first has a large spread, so only $\pm 1\sigma$ fits within the tolerance. The second has a spread three times smaller, so $\pm 3\sigma$ fits.



Tolerance Δ	Rods sold	Rods scrapped	Scrapped per million
1σ	68.3%	31.7%	317 000
3σ	99.73%	0.27%	2 700
6σ	99.9999998%	2×10^{-7} %	0.002 (2 per billion)

A process whose tolerance is $\pm 3\sigma$ is said to have **three sigma quality**: almost every rod can be sold. Large businesses extend the idea to **six sigma quality**, where the spread is so small that $\pm 6\sigma$ fits within the tolerance. (In industrial practice the mean is allowed to drift by up to 1.5σ over time, which raises the defect rate to the commonly quoted 3.4 per million.)

The trade-off. A smaller σ needs better machines, purer raw material and tighter control of the workshop, all of which cost money. Destroying rods also costs money: material, energy and machine time are lost. The manufacturer must balance the cost of upgrading the machines against the cost of scrapping rods. Six sigma quality pays only when a defect is very expensive or the production volume is very large.

The same idea in scientific discovery. Because of statistical uncertainty, a measurement repeated many times (for example, the number of counts in an energy window) does not give the same value every time; the results follow a distribution that the accepted theoretical model can predict. Here the logic runs the other way: instead of keeping results inside a tolerance, we ask whether an observation lies so far out in the tail of the predicted distribution that the model itself is in doubt. In particle and nuclear physics, a result more than 5σ above the prediction is taken as an indication that the model is not sufficient and needs to be corrected or extended. If the model were right, such a result would occur by chance only with probability 2.9×10^{-7}, about once in 3.5 million experiments (only the side of an excess counts, so this is one tail). A 3σ result is called only "evidence". Note that 2.9×10^{-7} is the probability of the observation *if the model is right*, not the probability that the model is wrong.

Example: the Higgs boson. At CERN's Large Hadron Collider, the model being tested was the known Standard Model processes *without* a Higgs boson, which predict a smooth background of events at each mass. In July 2012, the ATLAS and CMS experiments each found more events than this background near a mass of 125 GeV, mainly in decays into two photons and into four leptons. In each experiment the excess was about 5σ (5.9σ and 5.0σ in their papers), so the model without a Higgs could not explain it. Adding a new particle of mass 125 GeV, the Higgs boson predicted in 1964 by Higgs, Englert and Brout, corrected the model and explained the data. So the model found insufficient was the one without the Higgs; the Standard Model itself, which predicted the particle, was confirmed.

48 Exponential distribution

Memoryless processes. In Section 40 we described the decay of a nucleus as memoryless: whatever has happened so far, the probability of decay in the next short interval is proportional to the length of the interval. The same rule describes many other processes. A neutron travelling through a material has, in each short stretch δx of its path, the same probability $\Sigma\,\delta x$ of interacting with a nucleus, however far it has already travelled. A gamma ray behaves the same way, with probability $\mu\,\delta x$.

In general, let X be the time (or distance) until the event, and suppose

$$\Pr\big(\text{event in } (x, x+\delta x] \,\big|\, \text{no event up to } x\big) = \lambda\,\delta x.$$

Probability of the event between 0 and x. Let $S(x) = \Pr(X > x)$ be the probability that no event has happened up to x, with $S(0) = 1$. To have no event up to $x + \delta x$, there must be no event up to x *and* no event in the next δx:

$$S(x+\delta x) = S(x)\,(1 - \lambda\,\delta x) \quad \xrightarrow{\delta x \to 0} \quad \frac{dS}{dx} = -\lambda S \quad \Longrightarrow \quad S(x) = e^{-\lambda x}.$$

So the probability that the event happens between 0 and x is

$$F_X(x) = \Pr(0 < X \le x) = 1 - e^{-\lambda x}.$$

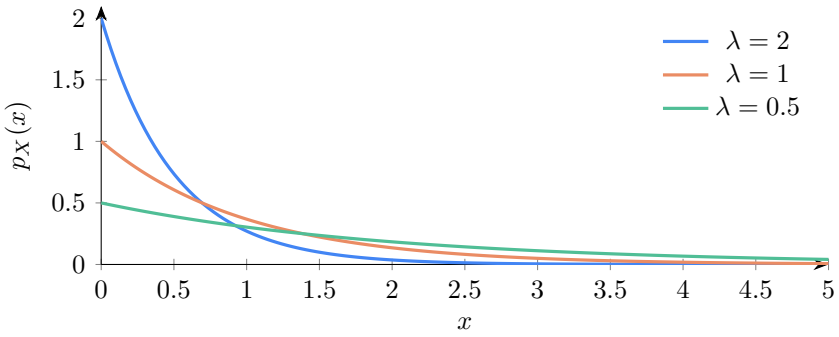
The PDF. For small δx, the probability that the event happens between x and $x + \delta x$ is (Section 43)

$$p_X(x)\,\delta x = F_X(x + \delta x) - F_X(x) = \frac{dF_X}{dx}\,\delta x = \lambda e^{-\lambda x}\,\delta x.$$

Exponential distribution. $X \sim \text{Exp}(\lambda)$ means

$$p_X(x) = \begin{cases} \lambda\, e^{-\lambda x} & x \geq 0, \\ 0 & x < 0, \end{cases}$$

with one parameter $\lambda > 0$.



What λ means.

Process	X	λ
Radioactive decay	lifetime of a nucleus (time)	decay constant (s^{-1})
Neutron interaction	distance travelled before interacting	macroscopic cross section Σ (cm^{-1})
Gamma-ray interaction	depth of penetration	attenuation coefficient μ (cm^{-1})

For a gamma beam, $S(x) = e^{-\mu x}$ is the fraction that passes through a thickness x without interacting: the familiar attenuation law $I = I_0 e^{-\mu x}$.

Decay constant and half-life. For radioactive decay, half of the nuclei have decayed by the half-life: $F_X(T_{1/2}) = \frac{1}{2}$. Hence

$$1 - e^{-\lambda T_{1/2}} = \frac{1}{2} \quad \Longrightarrow \quad \lambda = \frac{\ln 2}{T_{1/2}},$$

as promised in Section 40. The probability of decay within an observation time δt is $F_X(\delta t) = 1 - e^{-\lambda\,\delta t}$, which is approximately $\lambda\,\delta t$ when $\delta t \ll T_{1/2}$ (using $e^{-a} \approx 1 - a$ for small a).

Mean and variance

Start from the normalisation of the PDF:

$$\int_0^\infty \lambda e^{-\lambda x}\,dx = 1 \qquad \Longrightarrow \qquad \int_0^\infty e^{-\lambda x}\,dx = \frac{1}{\lambda}.$$

This holds for every $\lambda > 0$, so we may differentiate both sides with respect to λ. Each derivative brings down a factor $-x$, since $\frac{\partial}{\partial\lambda} e^{-\lambda x} = -x\, e^{-\lambda x}$:

$$-\int_0^\infty x\, e^{-\lambda x}\,dx = -\frac{1}{\lambda^2} \quad \Longrightarrow \quad \int_0^\infty x\, e^{-\lambda x}\,dx = \frac{1}{\lambda^2},$$

and differentiating this result once more,

$$-\int_0^\infty x^2 e^{-\lambda x}\,dx = -\frac{2}{\lambda^3} \quad\Longrightarrow\quad \int_0^\infty x^2 e^{-\lambda x}\,dx = \frac{2}{\lambda^3}.$$

Putting back the factor λ of the PDF:

$$\bar{X} = \int_0^\infty x\,\lambda e^{-\lambda x}\,dx = \lambda\cdot\frac{1}{\lambda^2} = \frac{1}{\lambda}, \qquad E[X^2] = \int_0^\infty x^2\,\lambda e^{-\lambda x}\,dx = \lambda\cdot\frac{2}{\lambda^3} = \frac{2}{\lambda^2},$$

$$\sigma_X^2 = E[X^2] - \bar{X}^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

For $X \sim \mathrm{Exp}(\lambda)$: $\bar{X} = \dfrac{1}{\lambda}$, $E[X^2] = \dfrac{2}{\lambda^2}$, $\sigma_X^2 = \dfrac{1}{\lambda^2}$, $\sigma_X = \dfrac{1}{\lambda}$.
The mean and the standard deviation of the exponential distribution are equal.

The mean $1/\lambda$ is the **mean lifetime** of a nucleus ($1/\lambda = T_{1/2}/\ln 2 \approx 1.44\,T_{1/2}$), the **mean free path** $1/\Sigma$ of a neutron, and the mean penetration depth $1/\mu$ of a gamma ray.

Radioactive decay: Poisson, exponential distribution and waiting time

Suppose events (decays, failures, arrivals) occur independently at a constant rate λ per unit time, so that the number of events in any interval of length δt is Poisson($\lambda\,\delta t$): the mean count is proportional to the length of the interval. Start a clock at any moment and let T be the waiting time until the next event.

The key observation: *the wait is longer than δt exactly when no event occurs in the interval* $(0, \delta t]$. So

$$\Pr(T > \delta t) = \Pr(\text{zero events in } (0,\delta t]) = \frac{(\lambda\,\delta t)^0 e^{-\lambda\,\delta t}}{0!} = e^{-\lambda\,\delta t}.$$

Therefore

$$F_T(\delta t) = \Pr(T \le \delta t) = 1 - e^{-\lambda\,\delta t}, \qquad p_T(\delta t) = \frac{dF_T}{d(\delta t)} = \lambda e^{-\lambda\,\delta t}, \qquad \delta t \ge 0,$$

which is exactly $T \sim \mathrm{Exp}(\lambda)$. Counts in fixed intervals are Poisson; the gaps between events are exponential with the same rate, and the mean gap is $1/\lambda$. For the gram of carbon-14 in Section 40, the rate of decays of the whole sample is its activity, $\lambda = A \approx 1.65 \times 10^{11}$ s^{-1} (the rate for the whole sample, not the decay constant of a single nucleus), so the mean time between successive decays is about 6×10^{-12} s.

Summary

Distribution	Values	$\Pr_X(x)$ or $p_X(x)$	Mean $\bar{X}$	Variance σ_X^2
$B(N,p)$	$0,1,\ldots,N$	$\binom{N}{x}p^x(1-p)^{N-x}$	Np	$Np(1-p)$
Poisson(λ)	$0,1,2,\ldots$	$\lambda^x e^{-\lambda}/x!$	λ	λ
$\mathcal{U}(a,b)$	$[a,b]$	$1/(b-a)$	$(a+b)/2$	$(b-a)^2/12$
$\mathcal{N}(\mu,\sigma^2)$	$(-\infty,\infty)$	$e^{-(x-\mu)^2/2\sigma^2}/\sqrt{2\pi\sigma^2}$	μ	σ^2
Exp(λ)	$[0,\infty)$	$\lambda e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$